

---

# ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

**Anonymous authors**  
Anonymous submission

## ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead whenever they process video, and this burden becomes especially acute for long or high-resolution inputs because of the massive number of visual tokens involved. To improve efficiency, a broad family of existing methods reduces redundancy by pruning or merging tokens according to importance or similarity. Yet these approaches, across their many variants, largely overlook a dimension of visual content that is central to virtually any temporal signal, i.e., changes and turning points, and they lack a unified model of spatio-temporal relationships. To address this, we advance a general perspective on token selection: similarity serves to identify redundancy, while difference serves to capture key events. Building on this principle, we design a training-free framework named ST-SimDiff that is applicable to a wide range of MLLM architectures without any model-specific adaptation. We first construct a spatio-temporal graph from the visual tokens to uniformly model their complex associations. Subsequently, we employ a parallel dual-selection strategy: 1) similarity-based selection uses community detection to retain representative tokens, compressing static information; 2) temporal difference-based selection precisely locates content-changing points to preserve tokens that capture key dynamic shifts. Together, these components preserve both static and dynamic content with a minimal number of tokens, and the design is general enough to accommodate diverse video types, task formats, and compression budgets. Extensive experiments across multiple benchmarks and backbone models show that our method consistently outperforms state-of-the-art approaches while substantially reducing computational costs. Our code will be released upon acceptance.

## 1 INTRODUCTION

The rise of Large Language Models (LLMs) has significantly propelled advancements of Large Vision-Language Models (LVLMs), which now demonstrate remarkable capabilities across a broad spectrum of image and video understanding tasks (Zhang et al., 2024c; Liu et al., 2024c). For video processing, LVLMs across current architectures typically sample a video into a sequence of frames and then convert each frame into hundreds or even thousands of visual tokens for the LLM (Bai et al., 2025; Li et al., 2024). While this paradigm is broadly effective, the number of tokens grows explosively with increasing video duration and resolution. This leads to prohibitive computational and storage burdens that recur regardless of the specific backbone, severely limiting the deployment of LVLMs across scenarios ranging from long-video analysis to real-time interaction (Fu et al., 2024b).

To address this challenge, a wide variety of methods have been proposed to enhance the efficiency of LVLMs, and they broadly fall into two families. One family is importance-based pruning. As observed by FastV (Chen et al., 2024), there is redundancy in the attention distribution across different layers when LVLMs process visual information. Consequently, visual tokens with lower attention scores in the deeper layers of the model can be pruned to reduce the computational load. The other family is similarity-based merging/selection, which builds on the general observation that visual tokens tend to be highly similar, whether between adjacent video frames or within a single frame.

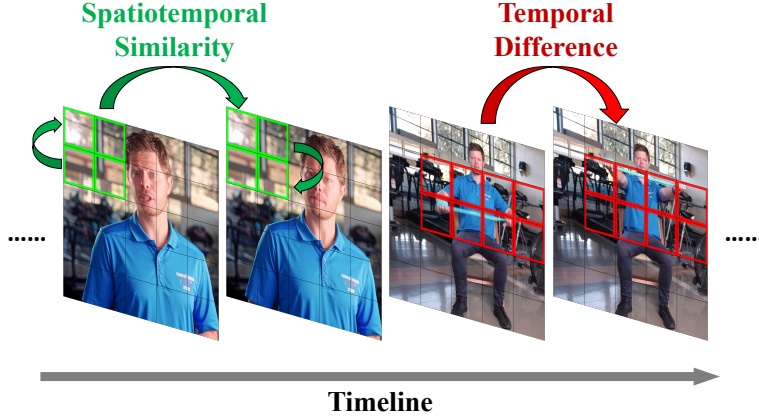


Figure 1: The core motivation of ST-SimDiff. We posit that efficient video understanding, across content types and tasks, requires handling two complementary scenarios simultaneously: Spatiotemporal Similarity (left) can be exploited to compress redundant information both spatially within a frame and temporally across adjacent frames. Temporal Difference (right) should be detected to capture the key actions or events that define the plot.

FrameFusion (Fu et al., 2024b) reduces redundancy by merging similar tokens from adjacent frames, while VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision encoder level.

Despite the broad success of existing methods in visual token reduction, they share two key limitations that persist across their variants. First, existing token pruning methods often focus on spatial correlations within the same frame or temporal correlations at the same positions across different frames. This lack of joint modeling and analysis of spatio-temporal correlations prevents them from effectively capturing the complex dynamic events that arise in diverse video content. Furthermore, these methods share a more fundamental blind spot: they mainly attend to informational commonalities, such as similarity and importance, while neglecting the crucial changes and differences that characterize temporal signals in general. Since the narrative of a video is typically driven by turning events, any model that only seeks similarity risks smoothing over a sudden action, leading to a misinterpretation of the content.

Based on this, we propose a general perspective on token selection: similarity is for identifying redundancy, while difference is for capturing key moments. We argue that an ideal video token compression algorithm, irrespective of the underlying content or backbone, should achieve two goals simultaneously: representing the stable content of a video with the fewest tokens, and precisely preserving the key changes. To this end, we propose **ST-SimDiff**, a dual token selection framework based on a spatio-temporal graph that is designed to generalize across models and video domains. Specifically, the framework consists of two parts. The first is **similarity-based representative token selection**: we treat all visual tokens of a video as nodes and construct a graph based on their feature similarity in the spatio-temporal dimension. Using a graph community detection algorithm, we find tightly connected token clusters in the graph, which represent stable and persistent visual elements in the video. We select a few central tokens with higher centrality from each cluster as representatives. The second part is **difference-based event token selection**: we pay special attention to the edges connecting along the time axis in the graph. When the similarity between corresponding tokens of adjacent frames drops sharply, we consider it a turning point and mark these tokens as critical event tokens that must be retained. Finally, we merge these two sets of tokens to be input into the LLM. Our contributions can be summarized as follows:

- We are the first to propose that the similarity and difference of tokens should be given equal importance in VLM efficiency research, framing this as a general principle rather than a benchmark-specific heuristic.
- We design a spatio-temporal graph framework to uniformly model the complex spatio-temporal relationships between video tokens. On this basis, we propose a novel dual token

selection strategy that simultaneously screens for both representative and pivotal event tokens, and that plugs into diverse LVLM backbones in a training-free manner.

- We conduct extensive experiments spanning multiple video understanding benchmarks and backbone models, and the results consistently show that ST-SimDiff significantly reduces the visual token budget while maintaining or even improving performance across settings.

## 2 RELATED WORK

### 2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has significantly propelled progress in multimodal understanding, giving rise to a broad and growing family of Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). Across their diverse designs, these LVLMs typically process images or video frames by converting them into visual tokens via pre-trained visual encoders, which are then fed into LLMs. A variety of alignment modules connect the visual encoder and the LLM, including MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023). This general architecture endows LVLMs with exceptional capabilities across a wide range of multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a). Video understanding is a particularly demanding instance, where they show strong promise in handling longer, higher-resolution, and more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

However, this powerful capability also introduces significant computational challenges that are especially pronounced in video understanding scenarios. Because a continuous sequence of frames must be processed, the number of visual tokens grows explosively, far exceeding that of static images, posing a severe challenge to the training and inference efficiency of LVLMs regardless of the specific model. Consequently, how to efficiently process massive video visual tokens and reduce computational overhead remains a broadly critical issue across current LVLM research.

### 2.2 VISUAL TOKEN COMPRESSION

Driven by the substantial overhead of video processing, visual token compression has emerged as a widely adopted approach for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods, taken as a whole, approach compression from two broad angles: token importance and similarity.

Earlier methods primarily focused on importance-based token pruning, aiming to identify and remove visual tokens that contribute less to model performance. These methods typically rely on attention scores or other feature metrics to quantify token importance, such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent works (Shang et al., 2024; Ye et al., 2025) have increasingly incorporated token similarity to more effectively reduce redundancy while preserving critical information, recognizing that even important tokens can exhibit high similarity and thus informational redundancy. For instance, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information, and VisPruner (Zhang et al., 2025b) leverages visual cues to remove redundant items.

Despite this broad progress on visual token redundancy, existing work rarely addresses the intricate, global spatio-temporal similarity relationships between tokens. In video scenarios of all kinds, visual information is redundant both spatially and temporally. Existing similarity-driven methods, such as FrameFusion, largely restrict themselves to the temporal similarity between adjacent frames, failing to exploit the richer spatio-temporal relationships available. This limits their ability to fully leverage the redundancy inherent to video and, in turn, constrains performance under aggressive token reduction across settings.

## 3 PROBLEM DEFINITION

Given a video  $V$  and a text query  $Q$ , a standard Large Vision-Language Model (LVLM), regardless of its particular backbone, first encodes the video into a full sequence of  $N$  visual tokens,  $T_{\text{full}} =$

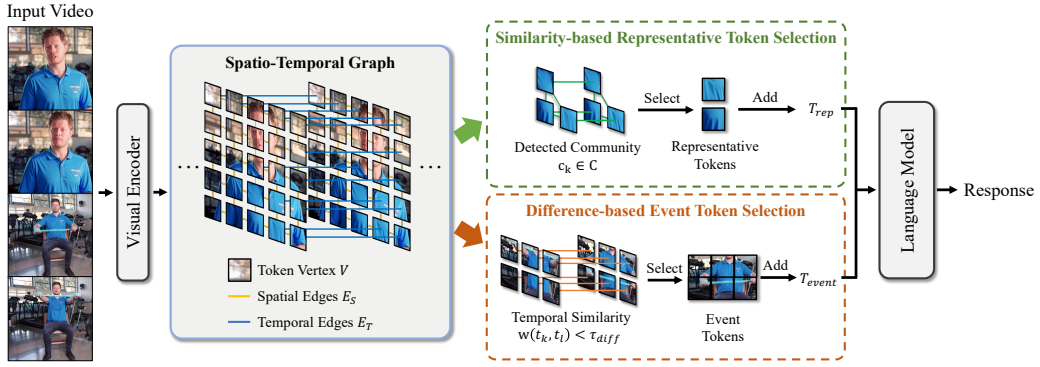


Figure 2: The overview framework of ST-SimDiff, which consists of three parts: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

$\{t_1, t_2, \dots, t_N\}$ . This process is computationally demanding, as the self-attention mechanism’s complexity scales quadratically with the token count  $N$ , i.e.,  $O(N^2)$ . Our task is therefore to design a general-purpose token selection function,  $f(\cdot)$ , that takes the full sequence  $T_{\text{full}}$  and a compression ratio  $r$  to produce a much smaller subset  $T_{\text{sub}} = f(T_{\text{full}}, r)$ . The objective is to maximize the LVLm’s downstream task performance using this subset, subject to the constraint that its size  $|T_{\text{sub}}|$  is approximately  $r \cdot N$ . Framed this way, the problem is model-agnostic and applies uniformly across compression budgets and task formats. The core challenge is to design the selection function  $f(\cdot)$  so that it retains the most critical information within the subset  $T_{\text{sub}}$ .

## 4 ST-SIMDIFF

### 4.1 METHOD OVERVIEW

As illustrated in Figure 2, the proposed ST-SimDiff method comprises two key components that operate in parallel on a spatio-temporal graph constructed from the video’s visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). Because both components act purely on token features and their spatio-temporal layout, the design carries over naturally to different encoders, backbones, and video domains. First, in the SRTS module, we identify and condense the video’s stable and redundant content. By applying community detection algorithms to the graph, we group highly similar tokens into clusters, which correspond to persistent visual elements such as static backgrounds. From each cluster, we then select a few highly central tokens to form a representative set,  $T_{\text{rep}}$ . Second, the DETS module is designed to capture the video’s crucial dynamic shifts. It analyzes the temporal edges of the graph, identifying moments where the similarity between corresponding tokens in adjacent frames drops sharply. These points of high difference signify turning points, and the tokens framing these transitions are preserved as critical event tokens, forming the set  $T_{\text{trans}}$ . By synergistically combining these two components, ST-SimDiff generates a final token subset  $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ , which is both highly compact and information-rich, retaining stable content while precisely capturing key events across a wide variety of videos.

### 4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder, whichever one the host model employs, transforms it into a sequence of tokens  $T = \{t_1, t_2, \dots, t_N\}$ . Each token  $t_k \in T$  corresponds to a specific spatio-temporal location and is associated with a feature vector  $\mathbf{x}_k$ . We define the location of each token  $t_k$  by its frame index  $T(t_k)$ , its spatial height index  $H(t_k)$ , and its spatial width index  $W(t_k)$ .

We model the relationships between these tokens using a spatio-temporal graph  $G = (V, E)$ , where the vertex set  $V$  represents all tokens in  $T$ . The edge set  $E$  is the union of two distinct subsets,  $E = E_S \cup E_T$ , which capture spatial and temporal relationships respectively. The spatial edges  $E_S$

connects tokens that are spatially adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges  $E_T$  connects tokens that are at the same spatial position but in adjacent frames, capturing temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

The weight of any edge  $w(v_i, v_j) \in E$  is defined by the cosine similarity of the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph structure efficiently encodes local spatial relationships and temporal continuity, providing a unified and general foundation that supports our dual-path token selection strategy irrespective of the specific model or video type.

#### 4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content of essentially every kind is characterized by substantial spatio-temporal redundancy. For instance, in a static scene that persists for several seconds, the background tokens exhibit high similarity over time. Similarly, for an object moving across the screen, its corresponding tokens, despite changing spatial positions, maintain a high degree of semantic correlation. Existing methods often address temporal and spatial similarities in isolation. In contrast, our spatio-temporal graph is designed to capture these joint similarities in a single unified structure. Within this graph, semantically related tokens form dense “communities” or “clusters” based on their feature similarity, irrespective of their specific frame or location. Selecting representatives from each community therefore emerges as a broadly efficient strategy for compressing redundant information while preserving core semantics.

To accurately identify strongly correlated token clusters, we first threshold the graph  $G$ . We set a similarity threshold  $\tau_{\text{sim}}$  and retain only the edges with weights above this threshold, forming a new graph  $G' = (V, E')$ , where  $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$ . Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on  $G'$  to identify token clusters  $C = \{c_1, c_2, \dots, c_m\}$ . For each community  $c_k \in C$ , we rank and filter its internal tokens based on their centrality. The centrality score  $S_c(t_a)$  of a token  $t_a \in c_k$  is defined as its average similarity with all other tokens within the community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we set the intra-community retention rate to the globally preset compression ratio  $r$ , and preserve the top  $\lceil |c_k| \cdot r \rceil$  tokens with the highest centrality scores from each community  $c_k$ . This uniform filtering process naturally handles all communities under any budget, including single-node communities composed of unique tokens (in which case the sole node is retained). Formally, the set of representative tokens,  $T_{\text{rep}}$ , is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of  $k$  elements with the highest scores from a given set. In summary, this strategy efficiently compresses the static and persistent content of a video by identifying and retaining the most central tokens within each semantic cluster, a mechanism that applies equally well to short clips and hours-long footage.

#### 4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, then difference defines its “events”. A model focusing only on similarity may excel at understanding “what is” but struggle with “what happened”. Across virtually all video domains, the plot is driven by key events, and the essence of an event is change—the appearance of a new object, the start of an action, or a scene transition. In our model, these changes manifest as a sharp difference in the features of temporally adjacent tokens. Accurately capturing these “turning points” of significant difference is therefore crucial for understanding dynamic content of any kind and correctly answering questions such as “when” and “why”.

Difference, particularly along the temporal dimension, often signals the occurrence of a key event. For example, the entry of a new object, a scene change, or the beginning/end of an action will cause drastic changes in the visual features at corresponding positions in adjacent frames. We specifically analyze the temporal edges ( $E_T$ ) in our spatio-temporal graph. For any temporal edge  $(v_k, v_l) \in E_T$  in the graph, which connects two temporally adjacent tokens  $t_k$  and  $t_l$ , we set a dynamic threshold  $\tau$  (e.g., the 95th percentile of all temporal edge difference scores). When the difference score of a temporal edge  $(v_k, v_l)$  exceeds this threshold, i.e.,  $w(t_k, t_l) < \tau_{\text{diff}}$ , we consider the subsequent token  $t_l$  as a critical event token and retain it. Formally, the set of event tokens,  $T_{\text{event}}$ , is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

With this strategy, tokens that signify moments of significant temporal change are reliably preserved. By capturing these key transitions, we ensure that the crucial dynamic aspects of a video’s narrative are not overlooked during compression, whatever the content or task at hand.

#### 4.5 OVERALL REDUCTION PROCESS

In the proposed framework, we first compute the representative token set  $T_{\text{rep}}$  and the event token set  $T_{\text{event}}$  in parallel, and take their union,  $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$ , as the initial candidate set. To precisely meet the target token count  $N_{\text{target}} = \lceil r \cdot N \rceil$ , we introduce a final pruning step. If the size of the candidate set,  $|T_{\text{candidate}}|$ , exceeds  $N_{\text{target}}$ , we remove the  $|T_{\text{candidate}}| - N_{\text{target}}$  tokens with the lowest importance. Following existing work (Chen et al., 2024), the importance of a token is determined by its attention score in the shallow layers of the LLM. This step allows our method to flexibly meet any computational budget while preserving critical information.

In summary, our overall method adeptly combines two general strategies: graph-based token selection and attention-based dynamic pruning. The former leverages the intrinsic structure of video content (similarity and difference) to ensure the retention of core information, while the latter provides a flexible mechanism to precisely meet any given computational budget. Together, this design makes our token compression process both principled and adaptive, achieving a robust balance between efficiency and performance that holds across models and operating points.

#### 4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of our proposed ST-SimDiff framework can be analyzed in three main stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The initial stage involves building the graph. For each of the  $N$  visual tokens, we compute the cosine similarity with its spatially and temporally adjacent neighbors. As each token has a small, constant number of neighbors, this step requires a single pass over all tokens, resulting in a complexity of  $O(Nd)$ , where  $d$  is the feature dimension of each token.

Following graph construction, the SRTS module identifies token clusters. As detailed in our implementation (Section 5.2), we employ a connected components algorithm, which operates with a near-linear time complexity of  $O(N + |E|)$ . Since the number of edges  $|E|$  is at most  $3N$ , this simplifies to  $O(N)$ . The subsequent filtering step within each community could be computationally intensive. A naive approach of calculating all-pairs similarity within each community  $c_k$  would have a complexity of  $O(|C| \cdot |c_k|^2 \cdot d)$ . To prevent this, we impose a constraint on the maximum size of any community. In the rare event that a community exceeds a predefined threshold (e.g.,

| Method                         | VideoMME    |             |             |             | LongVideoBench |             |             | EgoSchema   |
|--------------------------------|-------------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|
|                                | Short       | Medium      | Long        | Overall     | Relation       | Perception  | Overall     |             |
| Upper Bound (Full Performance) |             |             |             |             |                |             |             |             |
| LLaVA-Video                    | -           | -           | -           | 63.3        | -              | -           | 58.2        | 57.3        |
| Token Retain Ratio $r = 30\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 68.2        | 58.6        | 51.2        | 59.3        | 48.0           | 59.9        | 53.5        | 51.3        |
| PruMerge                       | 70.2        | 57.3        | 52.2        | 59.9        | 49.6           | 60.5        | 54.7        | 50.9        |
| FasterVLM                      | 71.3        | 57.6        | 51.4        | 60.1        | 51.0           | 61.3        | 55.8        | 52.6        |
| VisionZip                      | 68.4        | 55.9        | 50.4        | 58.3        | 46.9           | 60.3        | 53.2        | 53.0        |
| FrameFusion                    | 74.0        | 59.8        | 50.0        | 61.3        | 49.7           | <b>63.3</b> | 56.0        | 53.0        |
| <b>ST-SimDiff (Ours)</b>       | <b>74.7</b> | <b>61.9</b> | <b>53.0</b> | <b>63.2</b> | <b>52.5</b>    | 63.2        | <b>57.5</b> | <b>56.0</b> |
| Token Retain Ratio $r = 50\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 73.9        | 61.4        | 51.3        | 62.2        | 50.1           | 62.2        | 55.7        | 54.7        |
| PruMerge                       | 72.3        | 58.7        | 52.8        | 61.3        | 51.4           | 63.2        | 56.9        | 54.6        |
| FasterVLM                      | 74.0        | 59.2        | 51.8        | 61.7        | 51.3           | 62.2        | 56.4        | 56.2        |
| VisionZip                      | 72.6        | 57.7        | 51.6        | 60.6        | 51.3           | 63.2        | 56.8        | 54.2        |
| FrameFusion                    | 74.6        | 61.2        | 52.0        | 62.6        | 51.7           | 64.2        | 57.6        | 55.8        |
| <b>ST-SimDiff (Ours)</b>       | <b>76.2</b> | <b>62.4</b> | <b>52.7</b> | <b>63.8</b> | <b>52.3</b>    | <b>64.3</b> | <b>57.9</b> | <b>57.3</b> |

Table 1: Performance comparison across long-form video understanding benchmarks using LLaVA-Video-7B as the backbone with 64 input frames (%). The best performance among all methods is emphasized in **bold**.

$|c_k| > \sqrt{N}$ ), we partition it. This ensures the filtering process remains efficient and does not dominate the overall complexity. While other mainstream graph clustering algorithms such as the Louvain and Leiden methods offer more complex community definitions, their computational complexity is higher, approaching  $O(N \log N)$ . Therefore, we opted for the simpler and highly efficient connected components algorithm to ensure minimal processing overhead.

The DETS module involves a single pass through all temporal edges in the graph to identify significant changes by comparing token similarities against a threshold. This process has a linear complexity of  $O(Nd)$ .

In summary, the total complexity of our ST-SimDiff framework is governed by these linear-time operations, culminating in an overall complexity of  $O(Nd)$ . This is substantially more efficient than the quadratic complexity of the self-attention mechanism,  $O(N^2d)$ , used in the Large Language Model, and the gap only widens as inputs grow. As a result, our method provides significant token reduction with only a negligible computational footprint relative to the model’s main inference cost, a property that holds regardless of the backbone into which it is inserted.

## 5 EXPERIMENTS

### 5.1 EXPERIMENTAL SETUP

**Baselines** To evaluate our method’s effectiveness across settings, we adopt two representative base models spanning distinct architectures, LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c), and compare our approach against a broad range of state-of-the-art video token compression techniques. These baselines cover diverse strategies, including **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). To ensure a fair and complete comparison, we also include the original un-

| Method                         | VideoMME    |             |             |             | LongVideoBench |             |             | EgoSchema   |
|--------------------------------|-------------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|
|                                | Short       | Medium      | Long        | Overall     | Relation       | Perception  | Overall     |             |
| Upper Bound (Full Performance) |             |             |             |             |                |             |             |             |
| NVILA-Video                    | -           | -           | -           | 61.5        | -              | -           | 56.3        | 52.9        |
| Token Retain Ratio $r = 30\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 69.4        | 55.3        | 48.8        | 57.9        | 50.8           | 55.6        | 53.0        | 49.7        |
| PruMerge                       | 71.0        | 53.7        | <b>49.9</b> | 58.2        | 49.3           | 58.1        | 53.4        | 47.5        |
| FasterVLM                      | 72.6        | 56.9        | 51.0        | 60.1        | 48.7           | 57.8        | 53.0        | 49.3        |
| VisionZip                      | 71.4        | 56.2        | 49.7        | 59.1        | 46.3           | 56.2        | 50.9        | 48.9        |
| FrameFusion                    | 72.1        | 55.7        | 48.7        | 58.8        | 51.3           | <b>59.3</b> | 54.9        | 51.3        |
| <b>ST-SimDiff (Ours)</b>       | <b>73.3</b> | <b>57.7</b> | 49.7        | <b>60.2</b> | <b>51.7</b>    | 59.2        | <b>55.2</b> | <b>51.7</b> |
| Token Retain Ratio $r = 50\%$  |             |             |             |             |                |             |             |             |
| FastV                          | 71.9        | 58.3        | 49.3        | 58.9        | 49.5           | 58.8        | 53.9        | 50.2        |
| PruMerge                       | 71.2        | 54.8        | 46.8        | 57.6        | 49.7           | 58.7        | 53.9        | 48.9        |
| FasterVLM                      | 74.6        | 57.8        | 50.1        | 60.8        | 48.5           | 58.1        | 53.0        | 50.5        |
| VisionZip                      | 72.8        | 58.1        | 50.6        | 60.5        | 50.4           | 58.9        | 54.4        | 50.3        |
| FrameFusion                    | 72.7        | 57.2        | 48.3        | 59.4        | 51.0           | 59.2        | 54.8        | <b>52.6</b> |
| <b>ST-SimDiff (Ours)</b>       | <b>73.9</b> | <b>59.7</b> | <b>51.4</b> | <b>61.7</b> | <b>52.3</b>    | <b>61.3</b> | <b>56.5</b> | 52.5        |

Table 2: Performance comparison across long-form video understanding benchmarks using NVILA-Video-8B as the backbone with 64 input frames (%), complementing the LLaVA-Video results to probe generality across models. The best performance among all methods is emphasized in **bold**.

compressed model (Vanilla) as a performance upper bound and Random Sampling of an equivalent number of tokens as a lower bound.

**Benchmarks** To probe our method’s capabilities across the breadth of long-video understanding, we adopt three challenging and complementary benchmark datasets that together span a wide range of content types, video lengths, and reasoning skills: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME is a comprehensive benchmark for general video understanding, featuring diverse video types and three length categories: Short, Medium, and Long. LongVideoBench evaluates the ability to retrieve and reason about details in long videos through a “referring reasoning” task, spanning videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset focused on long-form, first-person videos that stresses the model’s ability to understand character intentions and causal chains of actions. Taken together, these benchmarks let us assess ST-SimDiff under markedly different domains and question formats rather than a single narrow setting.

## 5.2 IMPLEMENTATION DETAILS

We use NVIDIA L20 GPUs with 48GB VRAM on Ubuntu 22.04. The inference evaluation is conducted with the widely used *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow its original setting of 64 input frames, and to keep the comparison of computational efficiency and resource usage fair across models, we adopt the same 64-frame configuration for NVILA. For the hyperparameters, we set the similarity threshold  $\tau_{\text{sim}} = 0.8$  and the difference threshold  $\tau_{\text{diff}} = 0.2$ , and we deliberately keep this single configuration fixed across all models and benchmarks rather than tuning it per setting. Experiments span two representative operating points, with token compression rates of  $r = 30\%$  and  $r = 50\%$ . In our practical implementation, considering the trade-off between time consumption and performance improvement, we opt to use connected component finding as the community detection algorithm.

| Method                        | VideoMME    | LongVideoBench | EgoSchema   |
|-------------------------------|-------------|----------------|-------------|
| Token Retain Ratio $r = 30\%$ |             |                |             |
| Baseline                      | 60.3        | 56.2           | 54.8        |
| + Sim                         | Spatial     | 61.5           | 56.5        |
|                               | Temporal    | 61.7           | 56.8        |
|                               | Spa. + Tem. | 62.6           | 57.0        |
| <b>++ Diff (Proposed)</b>     | <b>63.2</b> | <b>57.5</b>    | <b>56.0</b> |
| Token Retain Ratio $r = 50\%$ |             |                |             |
| Baseline                      | 63.2        | 56.7           | 56.5        |
| + Sim                         | Spatial     | 63.3           | 57.0        |
|                               | Temporal    | 63.7           | 57.2        |
|                               | Spa. + Tem. | 63.7           | 57.8        |
| <b>++ Diff (Proposed)</b>     | <b>63.8</b> | <b>57.9</b>    | <b>57.3</b> |

Table 3: Ablation results of the different components across benchmarks (%).

### 5.3 COMPARISON WITH STATE-OF-THE-ARTS

To evaluate our proposed ST-SimDiff framework across a broad set of conditions, we compare it against a range of mainstream efficient video language model (VLM) compression methods. All experiments are conducted on three widely-used long-form video understanding benchmarks that together cover diverse domains and question types: VideoMME, LongVideoBench, and EgoSchema. To validate the generalization ability and robustness of our method, we report detailed performance data on two distinct base models, LLaVA-Video-7B and NVILA-8B, under token retain ratios of 30% and 50%. The experimental results are presented in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms all competing methods across every tested configuration, spanning both backbones, all three benchmarks, and both compression ratios. A particularly noteworthy finding is that at a 50% token retain ratio, the overall performance of our method on both base models not only surpasses other compression algorithms but, on some benchmarks, even matches or exceeds that of the original model using 100% of the tokens. This breadth of results validates the effectiveness and novelty of ST-SimDiff and motivates a deeper analysis of the strategies and limitations of existing baselines.

(2) Among importance-based baselines (e.g., FastV, FasterVLM), FasterVLM typically exhibits relatively stronger performance. However, a limitation common to this whole family is the inefficient handling of the temporal redundancy pervasive in videos, as they tend to retain important yet repetitive tokens, thereby limiting information compression efficiency. Hybrid baselines (e.g., PruMerge, VisionZip, FrameFusion) are more competitive than importance-only approaches, with FrameFusion generally showing the strongest overall results. Nevertheless, the shared optimization goal across these methods is still to identify and preserve representative “commonality” information, leaving a general blind spot: the tendency to overlook key narrative information driven by “turning points” and “abrupt events”.

(3) In summary, existing baselines collectively reflect a deepening understanding of “redundancy” in videos, yet they still face two core challenges that cut across methods: first, a lack of synergistic analysis of spatio-temporal joint similarity between visual tokens, which makes complex redundancies difficult to capture; and second, a general neglect of “difference” detection, which fails to actively preserve the key changes that define plot development. The consistent performance gain of ST-SimDiff stems from addressing both issues in a single unified framework by constructing a spatio-temporal graph to model complex spatio-temporal relationships and by introducing difference detection.

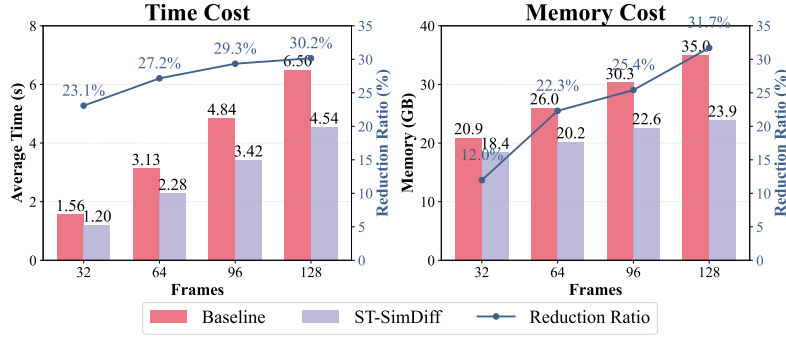


Figure 3: Computational cost comparison between our method and the baseline LLaVA-Video model, reporting average inference time and peak GPU memory usage under a 30% token budget across a range of input lengths.

#### 5.4 ABLATION STUDY

To validate the effectiveness of each core component within our ST-SimDiff framework, we conducted a series of detailed ablation studies. We started with a Baseline strategy that includes only fundamental importance-based pruning and progressively introduced our proposed similarity selection module (+Sim) and difference selection module (++Diff) to quantify their respective contributions. All experiments were conducted at token retain ratios of 30% and 50% and across the three benchmarks, with the results presented in Table 5.

(1) First, we evaluated the effectiveness of the +Sim module. The results consistently indicate that introducing similarity modeling on top of the baseline improves model performance across benchmarks. We further broke the similarity strategy down into three variants: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). The data show that collaboratively modeling the spatio-temporal associations between visual tokens is broadly the most effective way to compress redundancy.

(2) After integrating the optimal spatio-temporal joint similarity module, we introduced the difference selection module (++Diff) to form our complete framework. Adding this module yields a substantial performance leap, particularly under the aggressive 30% compression ratio. This highlights its vital complementary role: while the +Sim module focuses on redundancy, the ++Diff module acts as a safety net that identifies and preserves the unique event tokens defining the video’s narrative. Its marginal gain is smaller at the 50% retention ratio, since the more lenient +Sim module is more likely to have already captured these event tokens and overall performance is already approaching its upper bound. This analysis confirms our core motivation and its general applicability: the framework’s strength lies in balancing similarity-based redundancy compression with the crucial preservation of difference-based events, an advantage that is most pronounced in demanding high-compression scenarios but holds throughout.

#### 5.5 COMPUTATIONAL COST ANALYSIS

To quantify the efficiency advantages of our proposed ST-SimDiff method under realistic deployment conditions, we conducted detailed tests of its inference time and peak GPU memory usage under a 30% token budget, comparing against the baseline model (Baseline LLaVA-Video) without any compression. To characterize behavior across input scales rather than at a single point, the experiments in Figure 3 sweep video inputs from 32 to 128 frames.

**Inference Time** ST-SimDiff delivers a substantial and consistent speedup. As the number of input video frames increases, the average inference time of the baseline model grows sharply from 1.56 seconds (32 frames) to 6.50 seconds (128 frames). In contrast, our method effectively curtails this growth, requiring only 4.54 seconds on average when processing 128 frames. Notably, the time-saving rate scales up with video length, improving from 23.0% at 32 frames to 30.2% at 128 frames,

which indicates that the benefit becomes more pronounced precisely in the long-input regimes where efficiency matters most.

**Memory Cost** ST-SimDiff performs equally well on memory. The peak GPU memory usage of the baseline model climbs linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames). By substantially reducing the number of tokens fed into the large language model, our method markedly relieves this memory pressure: when processing 128-frame videos, ST-SimDiff’s peak memory usage is only 23.9 GB, a 31.7% saving relative to the baseline. This headroom enables the model to process longer videos on memory-constrained hardware, broadening the range of platforms and applications it can support.

In summary, ST-SimDiff not only achieves leading accuracy but also offers clear advantages in computational efficiency along both the time and memory axes, underscoring its practicality as a general solution for efficient video understanding.

## 6 CONCLUSION

In this paper, we address the computational inefficiency that Multimodal Large Language Models face when processing long videos, a problem rooted in the excessive number of visual tokens and the neglect of critical content changes that arises across models and content types. We propose ST-SimDiff, a training-free framework that builds a spatiotemporal graph to model complex associations between tokens. The method employs a dual selection strategy in which similarity-based selection identifies redundancy via community detection and temporal difference-based selection preserves tokens at key transition points. Extensive experiments across multiple benchmarks and backbone models show that ST-SimDiff substantially reduces processing time and memory usage while consistently surpassing state-of-the-art performance, and that these gains hold across the diverse settings we study. More broadly, this work opens a general research direction: by balancing content representativeness with transitional events, it points toward more efficient representations of video, and potentially of other temporal signals, that are not tied to any single model or benchmark.

## REFERENCES

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 1(8), 2023.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of statistical mechanics: theory and experiment*, 2008 (10):P10008, 2008.
- Liang Chen, Haozhe Zhao, Tianyu Liu, Shuai Bai, Junyang Lin, Chang Zhou, and Baobao Chang. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision*, pp. 19–35. Springer, 2024.
- Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning, 2023.
- Chaoyou Fu, Yuhang Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024a.

- Tianyu Fu, Tengxuan Liu, Qinghao Han, Guohao Dai, Shengen Yan, Huazhong Yang, Xuefei Ning, and Yu Wang. Framefusion: Combining similarity and importance for video token reduction on large visual language models. *arXiv preprint arXiv:2501.01986*, 2024b.
- Holdaway et al. Meta-designing quantum experiments with language models. *Nature Machine Intelligence*, 2024. doi: 10.1038/s42256-025-01153-0.
- Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26296–26306, 2024a.
- Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, January 2024b. URL <https://llava-vl.github.io/blog/2024-01-30-llava-next/>.
- Zhijian Liu, Ligeng Zhu, Baifeng Shi, Zhuoyang Zhang, Yuming Lou, Shang Yang, Haocheng Xi, Shiyi Cao, Yuxian Gu, Dacheng Li, et al. Nvila: Efficient frontier visual language models. *arXiv preprint arXiv:2412.04468*, 2024c.
- Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *Advances in Neural Information Processing Systems*, 36, 2024.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35: 27730–27744, 2022.
- Yuzhang Shang, Mu Cai, Bingxin Xu, Yong Jae Lee, and Yan Yan. Llava-prumerge: Adaptive token reduction for efficient large multimodal models. *arXiv preprint arXiv:2403.15388*, 2024.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *Advances in Neural Information Processing Systems*, 37:28828–28857, 2025.
- Haoning Xi, Zhiqi Shao, David A Hensher, John D Nelson, Huaming Chen, and Kasun Wijayaratna. A multi-task transformer with mixture-of-experts for personalized periodic predictions of individual travel behavior in multimodal public transport. *Transportation Research Part C: Emerging Technologies*, 179:105287, 2025.
- Senqiao Yang, Yukang Chen, Zhuotao Tian, Chengyao Wang, Jingyao Li, Bei Yu, and Jiaya Jia. Visionzip: Longer is better but not necessary in vision language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 19792–19802, 2025.
- Weihao Ye, Qiong Wu, Wenhao Lin, and Yiyi Zhou. Fit and prune: Fast and training-free visual token pruning for multi-modal large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 22128–22136, 2025.
- Zhang et al. Solving the many-electron schrödinger equation with a transformer-based framework. *Nature Communications*, 2025a. doi: 10.1038/s41467-025-63219-2.

---

648 Kaichen Zhang, Bo Li, Peiyuan Zhang, Fanyi Pu, Joshua Adrian Cahyono, Kairui Hu, Shuai Liu,  
649 Yuanhan Zhang, Jingkang Yang, Chunyuan Li, and Ziwei Liu. Lmms-eval: Reality check on  
650 the evaluation of large multimodal models, 2024a. URL [https://arxiv.org/abs/2407.](https://arxiv.org/abs/2407.12772)  
651 12772.

652 Qizhe Zhang, Aosong Cheng, Ming Lu, Zhiyong Zhuo, Minqi Wang, Jiajun Cao, Shaobo Guo,  
653 Qi She, and Shanghang Zhang. [cls] attention is all you need for training-free visual token prun-  
654 ing: Make vlm inference faster. *arXiv e-prints*, pp. arXiv-2412, 2024b.

655 Qizhe Zhang, Aosong Cheng, Ming Lu, Renrui Zhang, Zhiyong Zhuo, Jiajun Cao, Shaobo Guo,  
656 Qi She, and Shanghang Zhang. Beyond text-visual attention: Exploiting visual cues for effective  
657 token pruning in vlms. *arXiv preprint arXiv:2412.01818*, 2025b.

658 Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video  
659 instruction tuning with synthetic data. *arXiv preprint arXiv:2410.02713*, 2024c.

660  
661  
662  
663  
664  
665  
666  
667  
668  
669  
670  
671  
672  
673  
674  
675  
676  
677  
678  
679  
680  
681  
682  
683  
684  
685  
686  
687  
688  
689  
690  
691  
692  
693  
694  
695  
696  
697  
698  
699  
700  
701

## A VISUALIZATION OF THE TOKEN SELECTION PROCESS

To provide a more intuitive understanding of the ST-SimDiff framework, we present representative visualization samples of the intermediate and final token selection results in Figure 4. The figure illustrates the process on a video sequence featuring a dynamic object manipulation task, which serves as a concrete instance of the broader behaviors the framework exhibits. First, regarding the Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show how our graph community detection algorithm functions. It successfully groups spatially and temporally redundant regions, such as the static background and stationary objects, into cohesive communities. This confirms that SRTS reliably identifies stable visual elements and compresses them by selecting only a few central “representative tokens,” which appear as sparse yellow boxes in the final results, a mechanism that generalizes to any scene dominated by background redundancy.

Complementing this, the Difference-based Event Token Selection (DETS) plays a critical role during dynamic moments. As seen in the “Final Results” row, tokens triggered by high temporal difference are explicitly marked with dense red bounding boxes. These red boxes align closely with the moving hand and the shifting blue liquid, verifying that our method precisely locates “turning points” where the temporal difference exceeds the threshold ( $\tau_{diff}$ ). This ensures that fine-grained action details and rapid visual changes are preserved wherever they occur, allowing the model to capture the essence of an event without being overwhelmed by static data.

Finally, the synergistic effect of these two components is evident in the overlay of the final results. The visualization illustrates a general balance: the sparse yellow tokens represent stable background, while the dense red tokens capture rapid motion. This contrast provides direct visual evidence for our core motivation. ST-SimDiff avoids the blind spots of similarity-only approaches by densely sampling during rapid changes, while simultaneously pruning the spatiotemporal redundancies that importance-based methods often miss. Consequently, and independently of the particular clip, our framework strikes a favorable balance between computational efficiency and the preservation of critical visual information.

## B COMPARISON RESULTS ON QWEN2.5-VL

| Method                         | VideoMME |        |      |         | LongVideoBench |            |         | EgoSchema |
|--------------------------------|----------|--------|------|---------|----------------|------------|---------|-----------|
|                                | Short    | Medium | Long | Overall | Relation       | Perception | Overall |           |
| Upper Bound (Full Performance) |          |        |      |         |                |            |         |           |
| Qwen2.5-VL-64                  | 74.2     | 62.7   | 51.8 | 62.9    | 54.2           | 64.8       | 59.2    | 63.0      |
| Qwen2.5-VL-128                 | 77.4     | 65.8   | 56.1 | 66.4    | 56.2           | 65.1       | 60.4    | 63.9      |
| Token Retain Ratio $r = 30\%$  |          |        |      |         |                |            |         |           |
| ST-SimDiff-64                  | 73.8     | 61.7   | 51.8 | 62.4    | 52.2           | 62.1       | 56.8    | 62.7      |
| ST-SimDiff-128                 | 75.4     | 64.7   | 55.2 | 65.1    | 53.1           | 65.0       | 58.6    | 64.4      |
| Token Retain Ratio $r = 50\%$  |          |        |      |         |                |            |         |           |
| ST-SimDiff-64                  | 74.2     | 62.1   | 52.2 | 62.9    | 54.2           | 63.2       | 58.4    | 62.7      |
| ST-SimDiff-128                 | 76.8     | 66.3   | 55.8 | 66.3    | 54.9           | 65.3       | 59.8    | 64.5      |

Table 4: Performance comparison across long-form video understanding benchmarks using Qwen2.5-VL-7B as an additional backbone, at both 64 and 128 input frames (%).

To further probe the generalization of our proposed framework beyond the two backbones used in the main paper, we conducted additional experiments on the recently released Qwen2.5-VL model. These evaluations span the VideoMME and LongVideoBench benchmarks and are intended to confirm that our method’s efficacy is not tied to a particular model architecture but transfers broadly across backbones.

The empirical results, presented in Table 4, show the strong performance of ST-SimDiff on the Qwen2.5-VL model. At a token retention ratio of 30%, our method reaches 62.4 on VideoMME and 56.8 on LongVideoBench. When the token budget is increased to a 50% retention ratio, performance rises to 62.9 on VideoMME and 58.4 on LongVideoBench. It is particularly noteworthy that, at the

50% ratio, the performance on VideoMME matches the upper bound achieved using all tokens, while the score on LongVideoBench comes very close to its full-performance counterpart. Together with the main-paper results, these findings reinforce that ST-SimDiff generalizes across model families rather than to any single one.

## C ABLATION STUDY RESULTS ON NVILA

| Method                        |             | VideoMME    | LongVideoBench | EgoSchema   |
|-------------------------------|-------------|-------------|----------------|-------------|
| Token Retain Ratio $r = 30\%$ |             |             |                |             |
| Baseline                      |             | 59.3        | 53.8           | 50.2        |
| + Sim                         | Spatial     | 59.5        | 54.5           | 50.9        |
|                               | Temporal    | 59.6        | 54.3           | 51.2        |
|                               | Spa. + Tem. | 59.9        | 54.6           | 51.3        |
| <b>++ Diff (Proposed)</b>     |             | <b>60.2</b> | <b>55.2</b>    | <b>51.7</b> |
| Token Retain Ratio $r = 50\%$ |             |             |                |             |
| Baseline                      |             | 60.9        | 55.3           | 51.5        |
| + Sim                         | Spatial     | 61.1        | 55.7           | 51.9        |
|                               | Temporal    | 61.3        | 55.6           | 51.8        |
|                               | Spa. + Tem. | 61.4        | 56.0           | 52.3        |
| <b>++ Diff (Proposed)</b>     |             | <b>61.7</b> | <b>56.5</b>    | <b>52.5</b> |

Table 5: Ablation results of the different components on the NVILA-Video-8B model across benchmarks (%).

To assess whether the contribution of each component generalizes beyond a single backbone, we also conducted a series of detailed ablation studies on the NVILA-Video-8B model (Liu et al., 2024c). The experimental setup mirrors the ablation study in the main paper: we start with a Baseline strategy that includes only fundamental importance-based pruning and progressively introduce the similarity selection module (+ Sim) and the difference selection module (++ Diff).

As shown in Table 5, the results once again confirm the effectiveness of our framework design, and the trends are highly consistent with those observed on LLaVA-Video-7B, indicating that the findings are not specific to one model. First, after introducing the similarity module (+ Sim) on top of the baseline, performance improves steadily at both the 30% and 50% token retain ratios. Within the similarity module, we compared three strategies: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). Spatio-temporal joint modeling (Spa. + Tem.) is consistently the most effective; for example, at  $r = 50\%$  the joint strategy reaches 61.4% on VideoMME, outperforming the spatial-only and temporal-only variants. This again underscores the value of collaboratively modeling spatio-temporal associations when identifying redundant information. After integrating the optimal spatio-temporal joint similarity module, we further introduced the dissimilarity selection module (++ Diff) to form the complete ST-SimDiff framework. The table shows that adding this module brings the largest performance gains. Observed on a second backbone, this result strengthens the case that the two core modules (similarity and dissimilarity selection) generalize across models, and it again validates our central idea of simultaneously handling redundancy and change.

## D ABLATION RESULTS OF $\tau_{sim}$ AND $\tau_{diff}$ ON LLaVA-VIDEO

In our framework, the similarity threshold  $\tau_{sim}$  and the difference threshold  $\tau_{diff}$  are the two key hyperparameters that respectively govern the selection of representative and transitional tokens. To investigate their impact, we conducted a series of ablation studies, with the results shown in Figure 6. Across the ranges tested, both parameters exhibit a similar rise-then-fall trend and remain robust over a broad interval. For  $\tau_{sim}$ , a value that is too low leads to imprecise community detection,

---

while a value that is too high can fragment the integrity of semantic clusters. For  $\tau_{diff}$ , a value that is too low introduces noise through over-sensitivity, whereas a value that is too high may cause the model to miss key events. The model achieves its best performance at  $\tau_{sim} = 0.8$  and  $\tau_{diff} = 0.2$ , and the wide robust region means this single configuration transfers cleanly; we therefore adopt it across all other experiments.

## E ABLATION RESULTS OF $\tau_{sim}$ AND $\tau_{diff}$ ON NVILA-VIDEO

To assess how robust our method’s key hyperparameters are beyond the main-paper backbone, we repeated the threshold ablations on the NVILA-Video-8B model. As shown in Fig. 6, we evaluated the impact of the similarity threshold,  $\tau_{sim}$ , and the difference threshold,  $\tau_{diff}$ , on performance across two benchmarks: VideoMME and LongVideoBench.

**Impact of Similarity Threshold  $\tau_{sim}$ .** We swept  $\tau_{sim}$  over the range  $[0.6, 0.95]$ . Performance is relatively insensitive to this threshold: on VideoMME it peaks at 60.2% at  $\tau_{sim} = 0.8$ , and on LongVideoBench it is likewise optimal at  $\tau_{sim} = 0.8$ , reaching 55.2%. Although performance dips slightly when the threshold is set too high or too low, the overall fluctuation is minimal, confirming that our method stays stable across a wide range of similarity thresholds and across both benchmarks.

**Impact of Difference Threshold  $\tau_{diff}$ .** We swept  $\tau_{diff}$  over the range  $[0.1, 0.4]$ . As with  $\tau_{sim}$ , performance is strongly robust to this choice: on both benchmarks the optimum sits around  $\tau_{diff} = 0.2$ , with VideoMME at 60.2% and LongVideoBench at 55.2%. This indicates that a difference threshold near 0.2 is broadly effective for capturing key events, maximizing performance while compressing tokens.

In summary, these ablation studies confirm that our method is robust to its key hyperparameters. The results support our choice of  $\tau_{sim} = 0.8$  and  $\tau_{diff} = 0.2$  in the main experiments, as these values yield consistently strong performance across different benchmarks and different base models.

864  
865  
866  
867  
868  
869  
870  
871  
872  
873  
874  
875  
876  
877  
878  
879  
880  
881  
882  
883  
884  
885  
886  
887  
888  
889  
890  
891  
892  
893  
894  
895  
896  
897  
898  
899  
900  
901  
902  
903  
904  
905  
906  
907  
908  
909  
910  
911  
912  
913  
914  
915  
916  
917

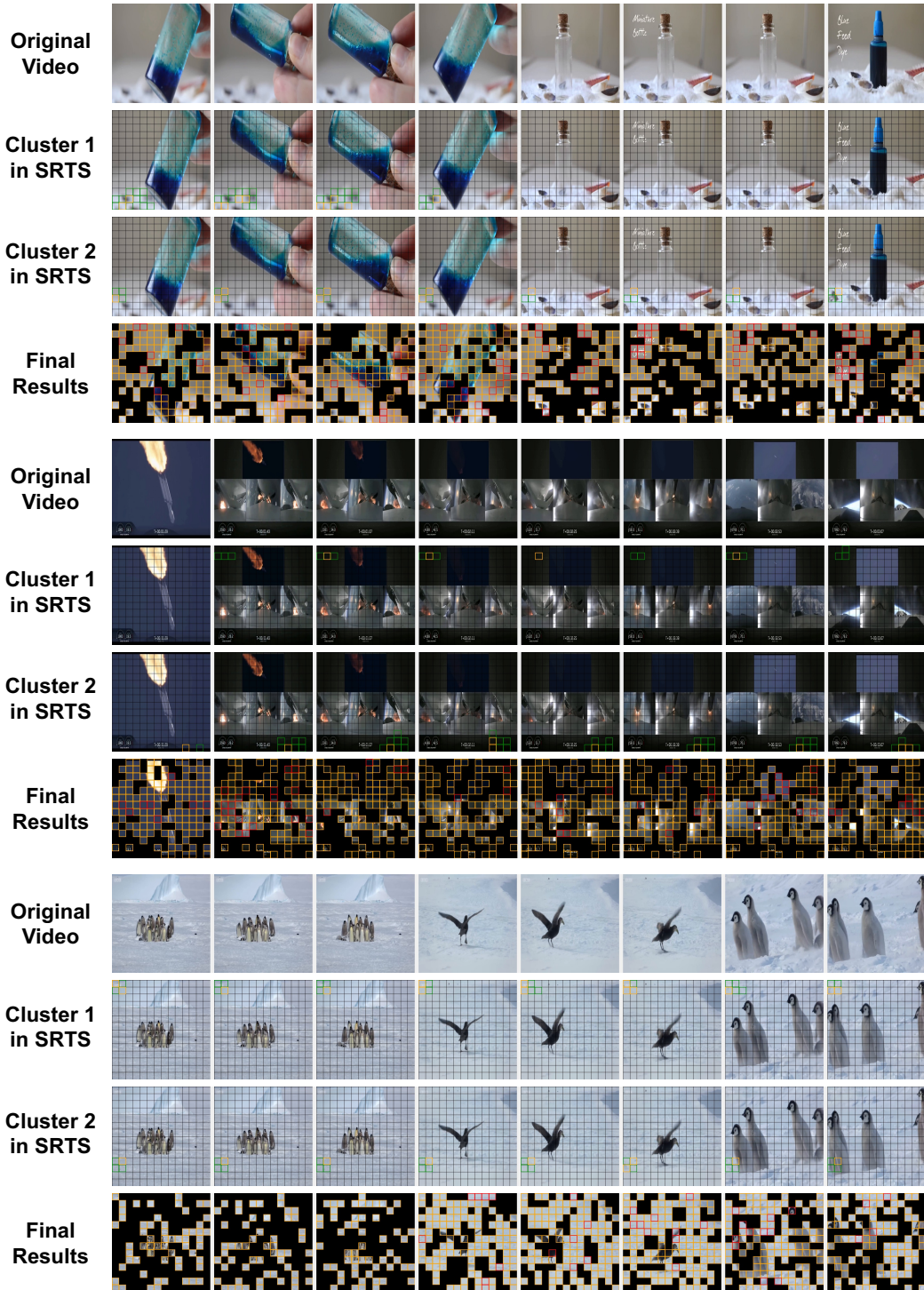


Figure 4: Visualization of the ST-SimDiff process on a representative example. The visualization breaks down token selection into communities detected by SRTS (shown as grids in Cluster 1 and 2) and event-driven tokens detected by DETS. The final result highlights the general synergy between sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

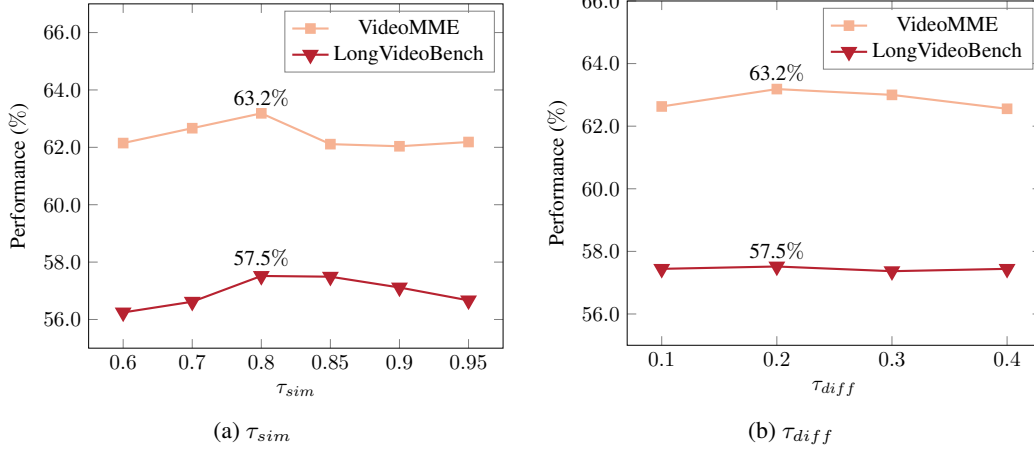


Figure 5: Ablation study results across different values of  $\tau_{diff}$  and  $\tau_{sim}$ , evaluated on both VideoMME and LongVideoBench.

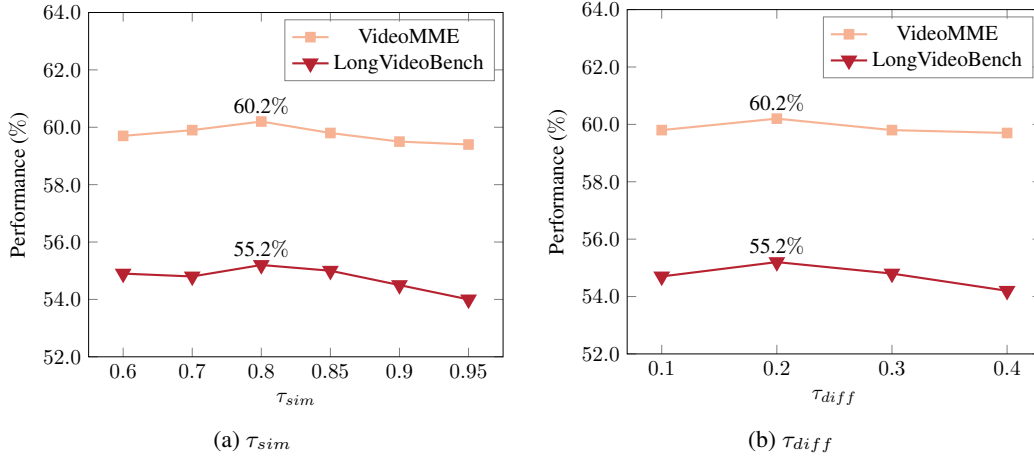


Figure 6: Ablation study results across different values of  $\tau_{diff}$  and  $\tau_{sim}$  on the NVILA-Video-8B backbone.